National Curriculum of Pakistan 2022-23

Assessment Framework

MATHEMATICS Grade-XI (HSSC-I)

Details of Content Area/ Domain/SLOs

Content Area/ Domain	Topics	SLO Code / Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks on SLOs	Number of Periods required (1 period = 40 minutes)
Domain A: Numbers and Algebra	Complex Numbers	[SLO: M-11-A-01]: Recall complex number z represented by an expression of the form $z = a + ib$ or of the form (a, b) where a and b are real numbers and $i = \sqrt{-\ }$.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	14
		[SLO: M-11-A-02]: Recognize a as a real part of z and b as an imaginary part of z.	Formative	U	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-11-A-03]: Know the condition for equality of complex numbers.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-11-A-04]: Carryout basic operations on complex numbers.	Formative	U	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-11-A-05]: Define $z = a - ib$ as the complex conjugate of $z = a + ib$.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	

		[SLO: M-11-A-06]: Define $\lvert z \rvert = \sqrt{a^2 + b^2}$ as the absolute value or modulus of a complex number $z = a + ib$.	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-A-07]: Solve the simultaneous linear equations with complex coefficients. For example, $\begin{cases} 5z - (3+i)w = 7 - i, \\ (2-i)z + 2iw = -1 + i \end{cases}$	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-08]: Write the polynomial (z) as a product of linear factors. For example, $\boldsymbol{z^2 + a^2 = (z + ia)(z - ia)}$ $\boldsymbol{z^3 - 3z^2 + z + 5 = (z + 1)(z - 2 - i)(z - 2 + i)}$	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-09]: Solve quadratic equation of the form $pz^2 + qz + r = 0$ by completing squares, where p, q, r are real numbers and z a complex number. For example, Solve $z^2 - 2z + 5 = 0$ $(z - 1 - 2i)(z - 1 + 2i) = 0$ $z = 1 + 2i, 1 - 2i$	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-10]: Explain the polar coordinates system.	**Formative**	U	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-A-11]: Describe the polar representation of a complex number.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-12]: Apply the operations with complex numbers in polar representation.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-13]: Demonstrate simple equations and in-equations involving complex numbers in polar form.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-14]: Apply concepts of complex numbers to real world problems (such as cryptography, wave phenomena, calculate voltage, current, circuits, the velocity and pressure of the fluid).	Summative	A	Question(s) will be asked in the annual examination.	

	Matrices & Determinants	**[SLO: M-11-A-15]: Apply matrix operations (addition / subtraction and multiplication of matrices) with real and complex entries.**	**Formative**	**K**	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	21
		[SLO: M-11-A-16]: Evaluate determinant of 3×3 matrices by using cofactors and properties of determinants.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-17]: Use row operations to find the inverse and the rank of a matrix.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-18]: Explain a consistent and inconsistent system of linear equations and demonstrate through examples.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-19]: Solve a system of 3 by 3 non homogeneous linear equations by using matrix inversion method and Cramer's Rule.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-20]: Solve a system of three homogeneous linear equations in three unknowns using the Gaussian elimination method.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-21]: Apply the concepts of matrices to real world problems such as (graphic design, data encryption, seismic analysis, cryptography, transformation of geometric shapes, social network analysis).	Summative	A	Question(s) will be asked in the annual examination.	
	Sequences and Series	**[SLO: M-11-A-22]:** Solve problems by analyzing arithmetic sequences and series up to n terms.	Summative	U	Question(s) will be asked in the annual examination.	21
		[SLO: M-11-A-23]: Solve problems by analyzing geometric sequences and series up to n terms.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-24]: Identify a sequence as arithmetic or geometric sequence up to n terms.	**Formative**	**U**	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	

		[SLO: M-11-A-25]: Solve problems by analyzing harmonic sequences and series up to n terms.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-26]: Find the sum of: **• first n natural numbers ($\sum n$),** **• squares of first n natural numbers ($\sum n^2$),** **• cubes of first n natural numbers ($\sum n^3$).**	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-A-27]: Recognize the arithmetic geometric sequence, determine its general term, find sum to n terms and sum to infinite number of terms, using sigma notation.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-28]: Identify leasing of motor vehicles, down payment, motor vehicle insurance, processing charges, repayment in monthly instalments.	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-A-29]: Solve problems related to leasing of motor vehicle under different conditions.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-30]: Apply concepts from sequence and series to real world problems (such as simple interest on loan, investment, depreciation, Investment planning on compound interest, projectile motion, gaming strategy, health care management, web page design, traffic modelling).	Summative	A	Question(s) will be asked in the annual examination.	
	Mathematical Induction and Binomial Theorem	**[SLO: M-11-A-31]: Describe a mathematical argument, identify the base case, induction of hypothesis and a precise conclusion.**	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	28
		[SLO: M-11-A-32]: Apply the principle of Mathematical Induction to prove statements, identities, divisibility of numbers and summation formulae.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-33]: Evaluate and justify conclusions, communicating a position clearly in an appropriate mathematical form in daily life.	Summative	A	Question(s) will be asked in the annual examination.	

		[SLO: M-11-A-34]: State and apply the Binomial Theorem to expand expressions of the form $(a+b)$ where n is a positive integer.	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-35]: Describe Binomial Theorem as expansion of binomial powers restricted to the set of natural numbers.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-36]: Calculate binomial coefficients using Pascal's triangle.	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-A-37]: Expand using the Binomial Theorem, and use appropriate techniques to simplify the expression.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-38]: Find an approximate value using the Binomial Theorem.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-39]: Use Binomial Theorem to find the remainder when a number to some large exponent is divided by a number.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-40]: Use Binomial Theorem to find the last digit of a number, test the divisibility by a number and compare two large numbers.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-41]: Apply concepts of Mathematical Induction and Binomial Theorem to real world problems such as (Puzzles, Domino Effects, Pascal's Triangle, Economic Forecasting, Rankings, Variable Subletting).	Summative	A	Question(s) will be asked in the annual examination.	
	Division of polynomial	**[SLO: M-11-A-42]:** Divide a polynomial of degree up to 4 by a linear and quadratic polynomial to identify quotient and remainder.	Summative	U	Question(s) will be asked in the annual examination.	14
		[SLO: M-11-A-43]: Demonstrate and apply Remainder Theorem.	Summative	U	Question(s) will be asked in the annual examination.	

		[SLO: M-11-A-44]: Analyze and apply Factor Theorem to factorize a cubic polynomial.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-A-45]: Apply concepts of Remainder and Factor Theorems to real world problems (such as polynomial regression, signal processing, and coding theory).	Summative	A	Question(s) will be asked in the annual examination.	
Domain B: Geometry	**Vectors in Space**	**[SLO: M-11-B-01]: Recognize rectangular coordinate system in space.**	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	28
		[SLO: M-11-B-02]: Define unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$ and Recognize components of a vector.	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-B-03]: Find the magnitude of a vector.	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-B-04]: Repeat all fundamental mathematical operations for vectors in space which, in the plane, have already been discussed.	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-B-05]: Demonstrate and prove properties of Vector Addition **• The Commutative Law,** **• The Associative Law,** **• Null vector $\vec{0}$ as an identity vector,** **• $-\vec{a}$ as additive inverse vector of $\vec{a}$.**	**Formative**	K	**Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.**	
		[SLO: M-11-B-06]: Explain Dot or Scalar Product of two vectors and give its geometrical interpretation.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-07]: Express dot product in terms of components.	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-08]: Find the condition for orthogonality of two vectors.	Summative	K	Question(s) will be asked in the annual examination.	

		[SLO: M-11-B-09]: Use dot product to find the angle between two vectors.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-10]: Find the projection of a vector along another vector.	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-11]: Find the work done by a constant force in moving an object along a given vector.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-12]: Solve daily life problems based on vectors. Cross or Vector Product.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-13]: Explain the cross or vector product of two vectors and give its geometrical interpretation.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-14]: Apply a cross product to find the angle between two vectors.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-15]: Solve situations in daily life based on Cross or dot Vector Product.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-16]: Describe scalar triple product of vectors.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-17]: Express Scalar Triple Product of vectors in terms of components (determinant form).	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-18]: Prove that • $\hat{i}\,\hat{j} \times \hat{k} = \hat{j}\,.\hat{k} \times \hat{i} = \hat{k}\,\hat{i} \times \hat{j} = 1$ • $\boldsymbol{j}\,.\boldsymbol{i} \times \boldsymbol{k} = \boldsymbol{k}\,.\boldsymbol{j} \times \boldsymbol{i} = \boldsymbol{i}\,.\boldsymbol{k} \times \boldsymbol{j} = -\mathbf{1}$. Prove that dot and cross are inter-changeable in scalar triple product.	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-19]: Find the volume of a • parallelepiped, • tetrahedron, determined by three given vectors.	Summative	K	Question(s) will be asked in the annual examination.	

		[SLO: M-11-B-20]: Define coplanar vectors and find the condition for planarity of three vectors.	Summative	K	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-21]: Apply concepts of vectors in space to real world problems such as (design and execute optimal navigation paths in transportation and logistics, graphing complex 3D motion, vector operations in engineering and computer graphics, practical proficiency for work, flux, and circulation).	Summative	A	Question(s) will be asked in the annual examination.	
	Fundamental Law of Trigonometry	**[SLO: M-11-B-22]:** Establish fundamental law of trigonometry.	Summative	U	Question(s) will be asked in the annual examination.	14
		[SLO: M-11-B-23]: Apply fundamental law and its deductions to derive: Trigonometric ratios of allied angles, double angle, half angle and triple angle identities.	Summative	A	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-24]: Express the product (of sines and cosines) as sums or differences (of sines and cosines).	Summative	K	Question(s) will be asked in the annual examination.	
	Trigonometric Functions	**[SLO: M-11-B-25]:** Find the domain and range of the trigonometric functions.	Summative	U	Question(s) will be asked in the annual examination.	21
		[SLO: M-11-B-26]: Discuss even, odd functions and the periodicity of trigonometric functions.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-27]: Find the maximum and minimum value of a given function of the type: • $a + bsin\theta$, • $a + b\ cos\theta$ • $a + bsin(c\theta + d\ )$, • $a + b\ cos\ cos\ (c\theta + d)$ • the reciprocals of above, where a, b, c and d are real numbers.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-28]: Graph and analyze the trigonometric functions sine, cosine, and tangent to solve problems.	Summative	U	Question(s) will be asked in the annual examination.	

		[SLO: M-11-B-29]: Explain the properties of graphs of $sin\theta, cos\theta$ and $tan\theta$.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-B-30]: Apply the concepts of trigonometric functions, identities, graphs, periodicity, even & odd functions and extreme values to real world problems such as distance, elevation, and direction of tall structures, navigation and mapping, lengths of irregular shapes, graphs to visualize and predict patterns in data, frequency and periodic length of Ferris wheel, forces on a see-saw or lever, the ideal angle for solar panel placement.	Summative	A	Question(s) will be asked in the annual examination.	
Domain C: Information Handling	**Permutation and Combination**	**[SLO: M-11-C -01]:** Explain and solve problems that involve the fundamental counting principle.	Summative	U	Question(s) will be asked in the annual examination.	14
		[SLO: M-11-C -02]: Explain and Solve problems that involve permutations.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-C -03]: Explain and Solve problems that involve combinations.	Summative	U	Question(s) will be asked in the annual examination.	
		[SLO: M-11-C -04]: Apply the concepts of permutation and combination to real world problems such as cryptography, estimating the odds of winning a lottery, calculating the number of possible DNA sequences or protein structures, choosing different sets of songs for certain occasions.	Summative	A	Question(s) will be asked in the annual examination.	